Data Structures: Stacks

A stack is a linear data structure that follows LIFO (Last In, First Out) order. Elements are added with push and removed with pop, and both happen only at one end, called the top. Peek returns the top element without removing it. Push, pop and peek each take O(1) time. A stack can be implemented with an array or a linked list. Popping from an empty stack causes underflow; pushing onto a full array-based stack causes overflow. Stacks are used for function calls (the call stack), undo in text editors, checking balanced parentheses, and converting infix expressions to postfix.